

孫正義育英財団

Generation 9

ADAM CHEONG

BORN IN 2009

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Masason Foundation



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**Class of 2027
HIGH SCHOOL**

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Education

2023.08 - 2027.06

High School

San Francisco, California

Proof School



For Kids Who Love Math

- 11th: Classic Literature: The Essay
- 11th: Medicinal Chemistry
- 11th: Anatomy & Physiology
- 11th: Quantum Mechanics
- 11th: Axiomatic Set Theory
- 11th: Linear Topology
- 11th: Linear Algebra A & B
- 11th: US History & the American Essay
- 10th: Literary Arts: Graphic Narrative
- 10th: Generating Functions
- 10th: Calculus A, B, C & D
- 10th: Physics
- 10th: Chemistry



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Adam Cheong

has earned the

AP[®] Scholar with Honor Award

which recognizes exemplary college-level achievement, for attaining an average score of at least 3.25 on all AP Exams taken, and scores of 3 or higher on four or more exams.

David Coleman, CEO, College Board

Trevor Packer, Senior Vice President, AP and Instruction

July 03, 2025

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Competition	AIME Threshold	Honor Roll Distinction	Distinction
AMC 10 A	105	136.5	112.5
AMC 12 A	96	150	127.5
AMC 10 B	99	133.5	105
AMC 12 B	100.5	145.5	127.5

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Berkeley Class Schedule

[Class Schedule](#)[Course Catalog](#)[Undergraduate](#)[Graduate](#)[Archive](#)

TERM

- ☒ Summer Sessions 2025 (1)
 - ☐ C: June 23-Aug 15 (1)
- ☐ Summer Sessions 2024 (1)
 - ☐ C: June 17-Aug. 9 (1)
- ☐ Spring 2024 (2)
- ☐ Fall 2023 (1)
- ☐ Summer Sessions 2023 (1)
 - ☐ C: June 20-Aug. 11 (1)
- ☐ Spring 2023 (2)
- ☐ Fall 2022 (1)

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MODE OF INSTRUCTION

- ☐ Online Instruction (1)

MAJOR REQUIREMENTS

--Select requirement year--

Search found 1 items

☒ Summer Sessions 2025

Search

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2025 Summer Session C 8 weeks, June 23 - August 15 **COMPSCI 61A**
001 - LEC 001 offered through **Electrical Engineering and Computer Sciences**
The Structure and Interpretation of Computer Programs

Laryn Qi

Jun 23, 2025 - Aug 15, 2025

Mo, Tu, We, Th

05:00 pm - 06:29 pm

Class #: 13253 Units: 4

Instruction Mode: Online

Time Conflict Enrollment Allowed

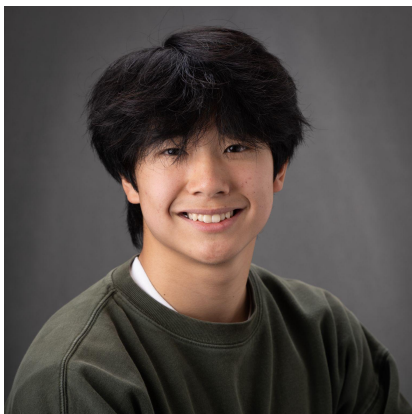
Open Seats

30 Unreserved Seats

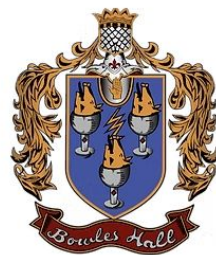
An introduction to programming and computer science focused on abstraction techniques as means to manage program complexity. Techniques include procedural abstraction; control abstraction using recursion, higher-order functions, generators, and streams; data abstraction using interfaces, objects, classes, and generic operators; and language abstraction using...

[How to apply](#)

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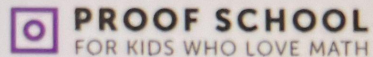


Berkeley
UNIVERSITY OF CALIFORNIA



BOWLES HALL
RESIDENTIAL COLLEGE





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Enrollment

Summer 2025

Class	Title	Un.	Gr.	Pts.
COMPSCI 61A (Session C)	The Structure and Interpretation of Computer Programs	4	A	16

Summer 2026

Class	Title	Un.	Gr.	Pts.
COMPSCI 61BL (Session C)	Data Structures and Programming Methodology	4	—	—
DATA C8 (Session C)	Foundations of Data Science	4	—	—

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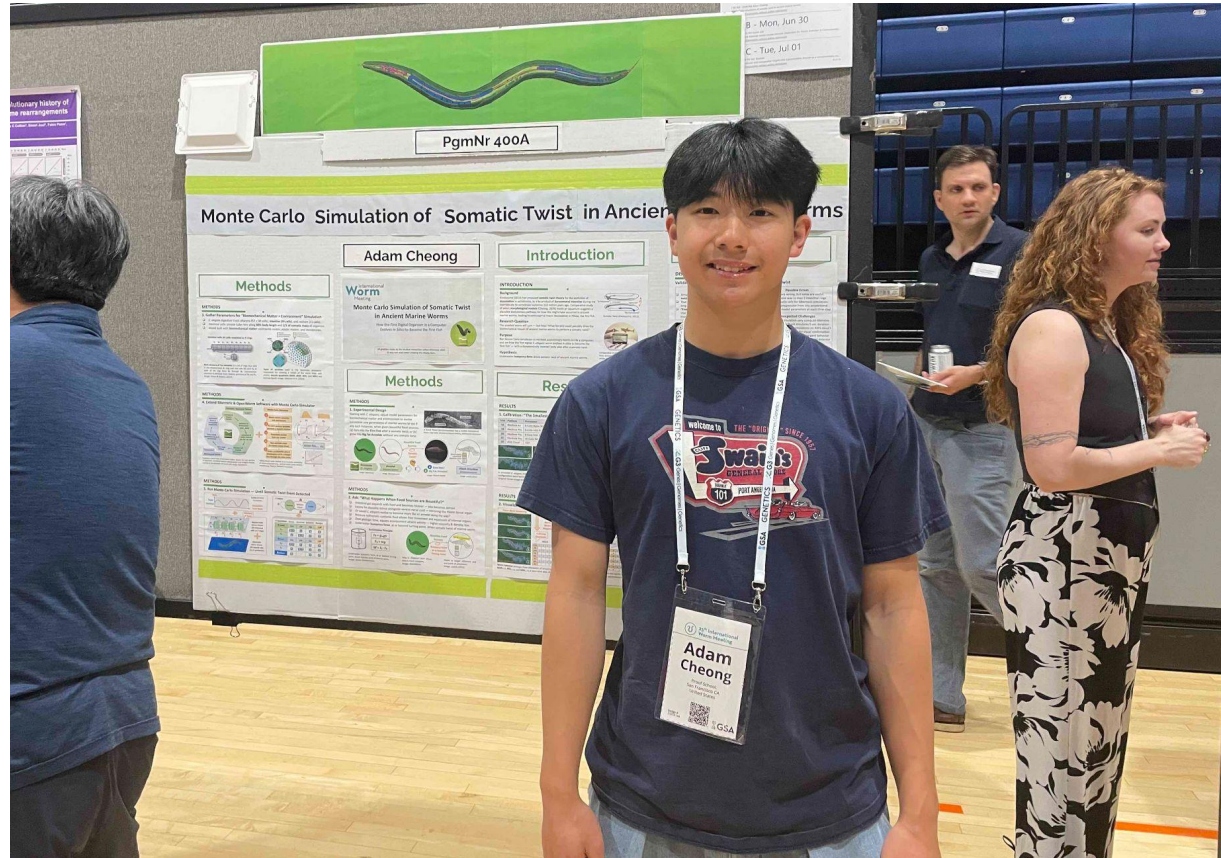
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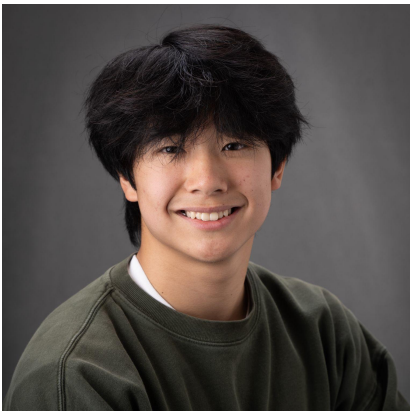




PEQG26

Population, Evolutionary and
Quantitative Genetics Conference





How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions

Adam Cheong^{1,2}, Mark Cheong², and Paul Fong³
¹Proof School, ²UC Berkeley, ³Duke University

Project Summary

The vast diversity of genetic and phenotypic characteristics observed in human cancers, according to Hanahan (2020), can be understood in the context of the cancer cells as a multicellular pathogen of immunogenetics that lack the formation of primary tumors, which subsequently evolve and progress due to selective advantage in the face of multifaceted barriers to controlling proliferative mass extinctions resulting in metastasis and adaptive resistance. Here we elucidate the adversarial response, a prescriptive framework borrowed from epidemiology (Kiers & West, 2008) for modeling therapeutic intervention targeted at the primary tumor, whose microenvironment undergoes remodeling and subsequently becomes unfit for cancer cells proliferation, illustrating how subpopulations of residual cancer cells evolve to survive multiple mass extinctions originating from a prescribed course of antibody drug conjugate targeting the primary tumor, potentially laying the seeds for future metastasis or adaptive resistance. We propose to conduct a simulation leading to further explore cancer population dynamics in the underexplored the mechanism of relapse.

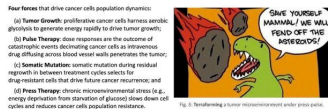
Introduction

Antibody-Drug Conjugates Show Superior Response when Dosed at MTD



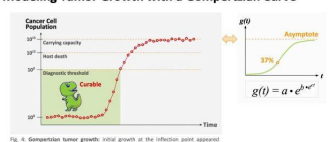
Research Question

How to Make a Tumor Microenvironment "Most Unfit" Under Press-Pulse?



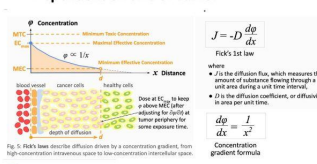
Methods — Growth

Modeling Tumor Growth with a Gompertzian Curve

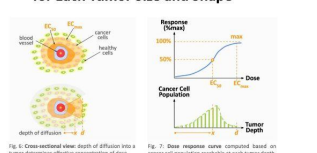


Methods — Pulse

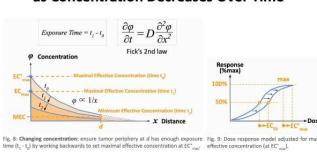
1 Dose at Maximal Effective Concentration — Dependent on Size of Tumor



2 Compute Dose Response Curve — for Each Tumor Size and Shape

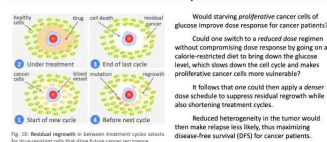


3 Adjust Dose Response Model — as Concentration Decreases Over Time



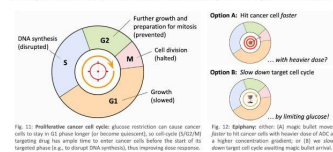
Discussion — Mutation

Cancer Cells Evolve in Between Treatment Cycles to Survive Mass Extinctions



Discussion — Press

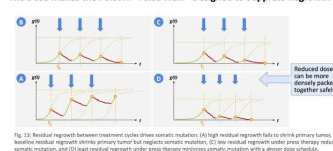
Making Proliferative Cancer Cells Vulnerable Through Bioenergetic Intervention



13 Realistic regrowth between treatment cycles depends on several factors: (A) high residual regrowth leads to shrink primary tumors, (B) moderate regrowth between primary tumor and metastatic lesions, (C) low residual regrowth between primary tumor and metastatic lesions, (D) low residual regrowth between primary tumor and metastatic lesions, (E) low residual regrowth between primary tumor and metastatic lesions.

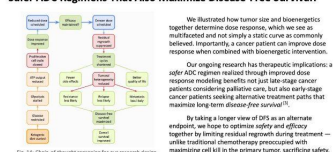
Conclusions

The Dose Makes the Poison: "Pulse Train" Designed to Suppress Regrowth



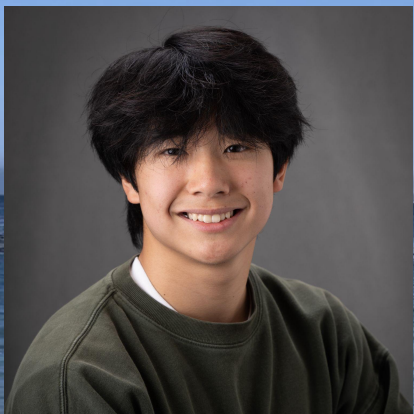
Future Work

Safer ADC Regimens That Also Maximize Disease-Free Survival?

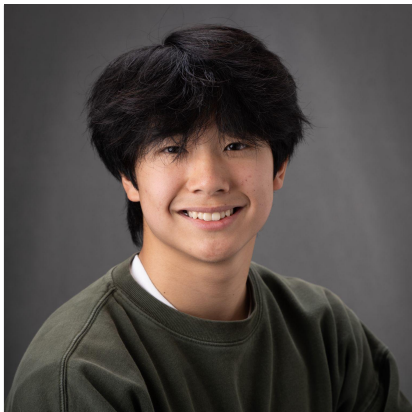


References

1. Arora, N. C., & West, D. (2020). Press-pulse: a general theory of mass extinction? *Paleobiology*, 46(4), 458–471.
2. Arora, N. C., & West, D. (2020). Objective response rate assessment in oncology: Current situation and future perspectives. *World Journal of Clinical Oncology*, 11(2), 53–73.
3. Duggan, A., & Gaudin, A. (2021). Clinical endpoints in oncology — a primer. *Annals of Cancer*, 31(1), 1120–1131.
4. Duggan, A., & Norton, L. (2021, September). Implication of Cancer Growth Kinetics for the Further Development of Antibody-Drug Conjugates. *CMAJ*, 193(18), 1611–1618.
5. Hanahan, D. (2020). Hallmarks of cancer — Then and now, and beyond. *Cell*, 180, 281–318.
6. Mooney, J., Detert, K., & Gaudin, A. (2023, July 4). Antibody Drug Conjugates: windows of opportunity. *Drug Target Review*.
7. Schick, K. A., Gallagher, K., Masai, M. A., Robertson-Tessi, M., Gaudin, C. A., West, J., Lian, P., Prohasko, S., Gaudin, A., & Anderson, A. R. (2023). Mathematical Oncology: How Modeling in Translational Clinical Decision-Making. *Cancer Research*, 83(24), 4868–4878.
8. Seifried, T. M., Yu, Y., Mooney, J. C., & Gaudin, A. (2023). Press-pulse: a novel therapeutic strategy for the metabolic management of cancer. *Anticancer Research*, 43(1).
9. Yoon, N., Boeckle, J., & Hymet, D. M. (2020). A new drug resistance in cancer. *Nature*, 575(7783), 299–308.







University of Oxford

Degree level: **Undergraduate**

BIOMEDICAL ENGINEERING

Course **options**

2026 - 2027

28 course options available



Qualification type

Master of Engineering (with Honours)

There are other course options available which may have a different vacancy status or entry requirements.



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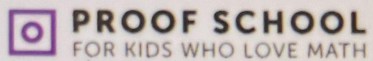
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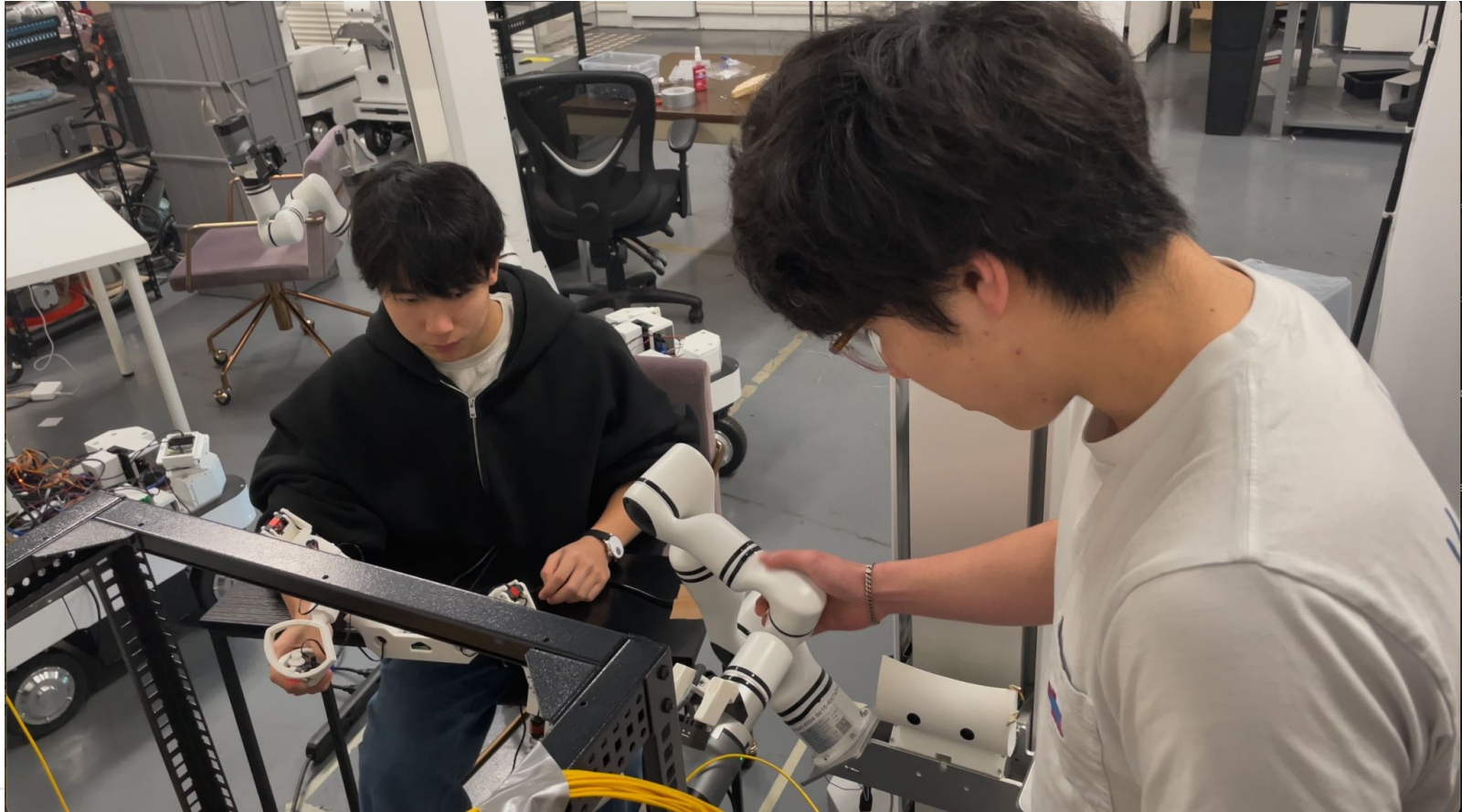
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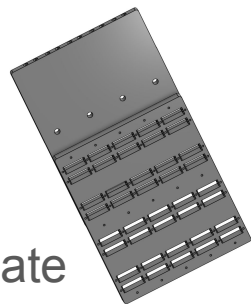
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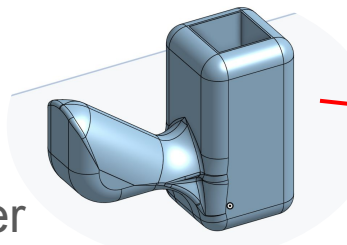




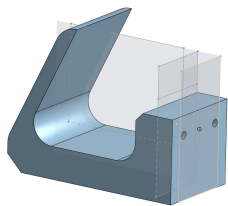
Face Plate



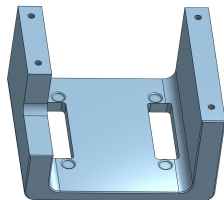
Antler



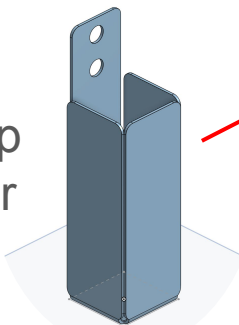
Wire Hook



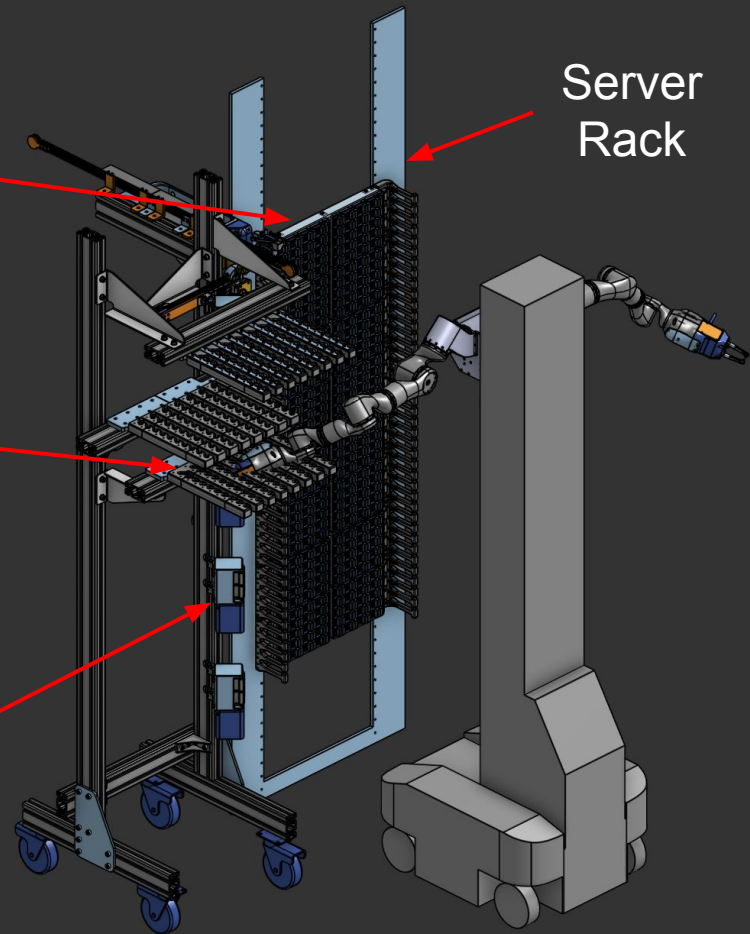
Wire Dock Holder



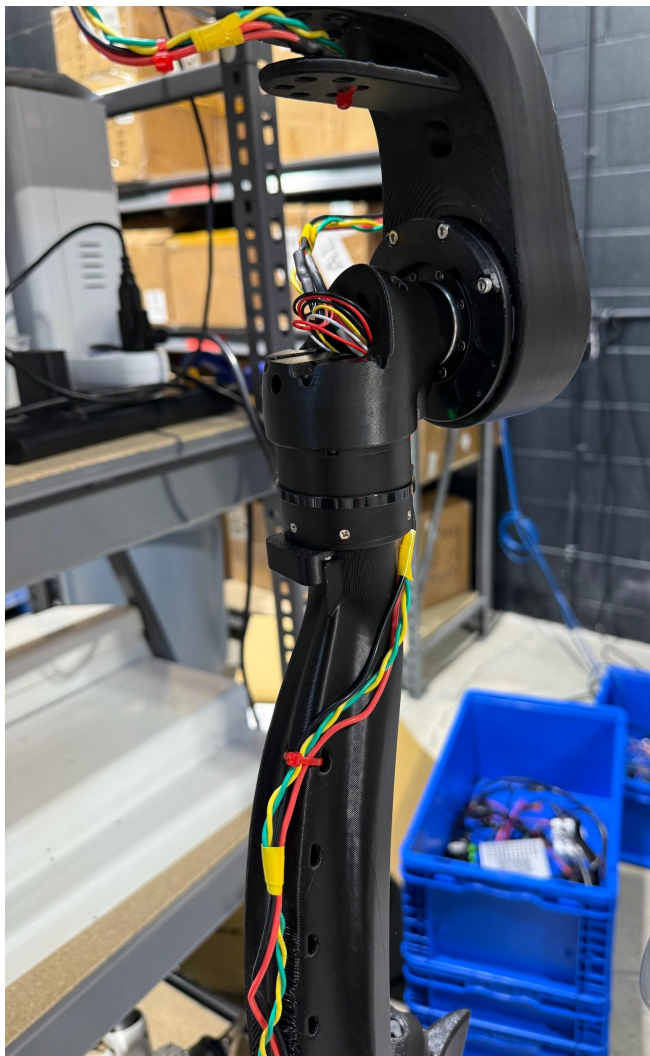
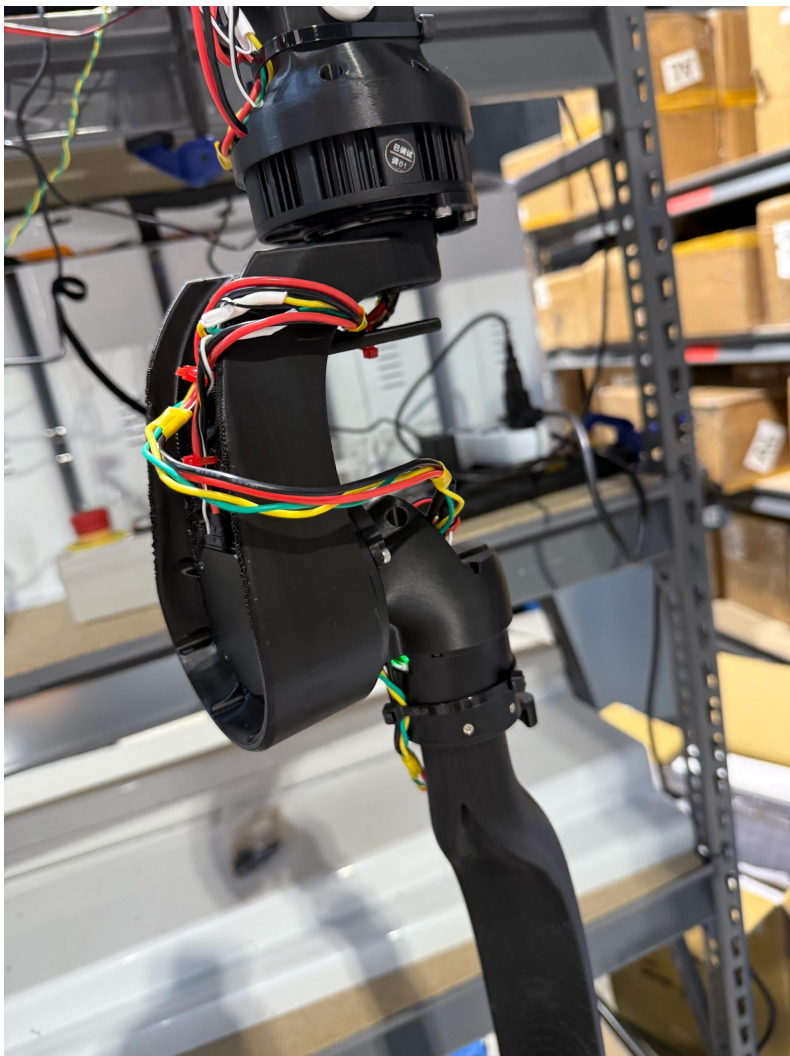
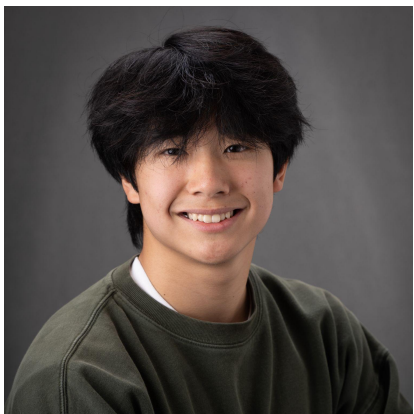
Wire Cap Collector

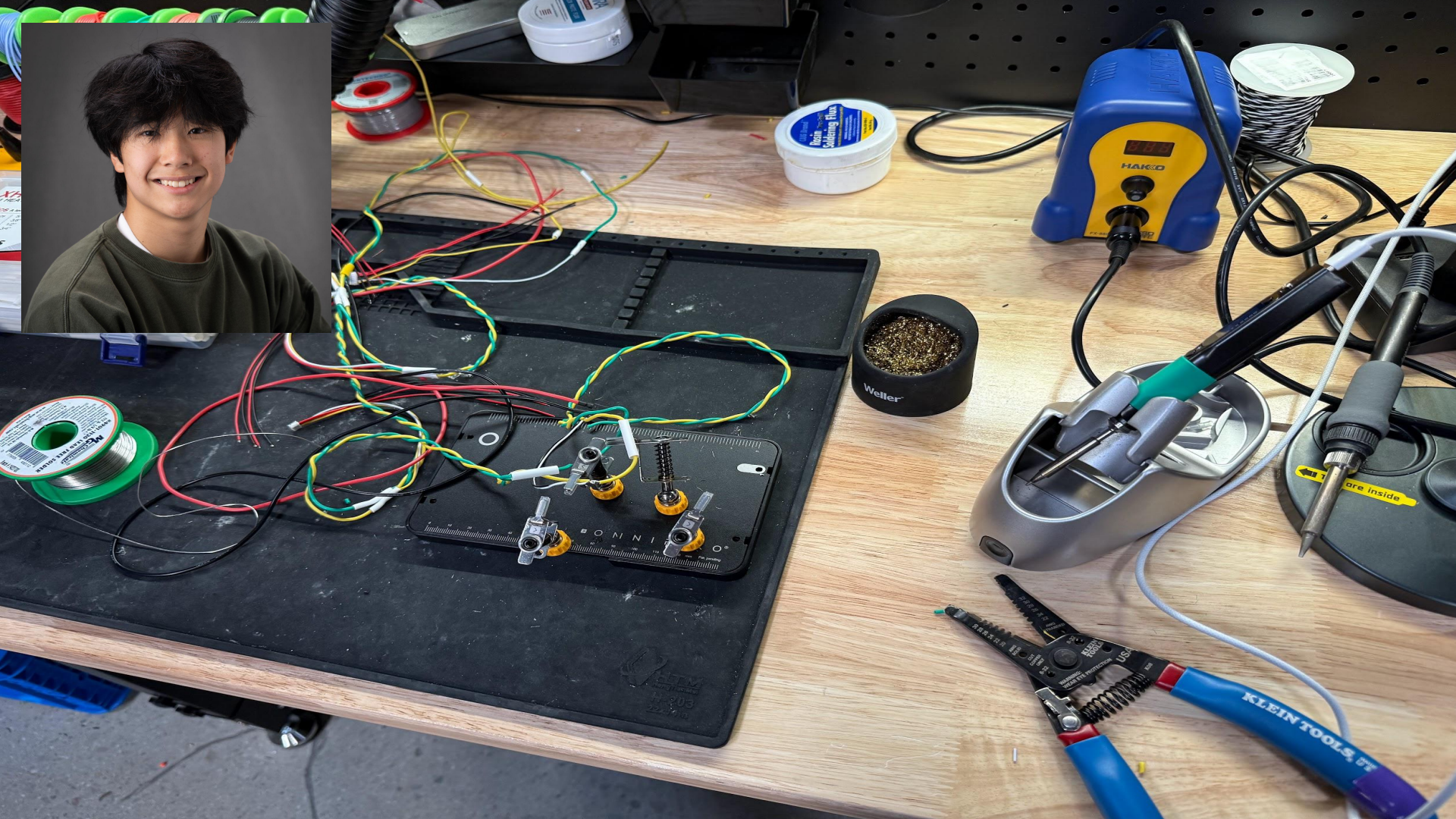
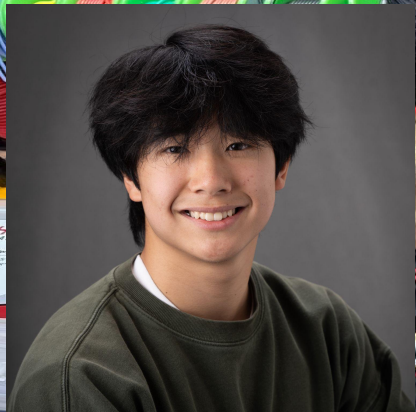


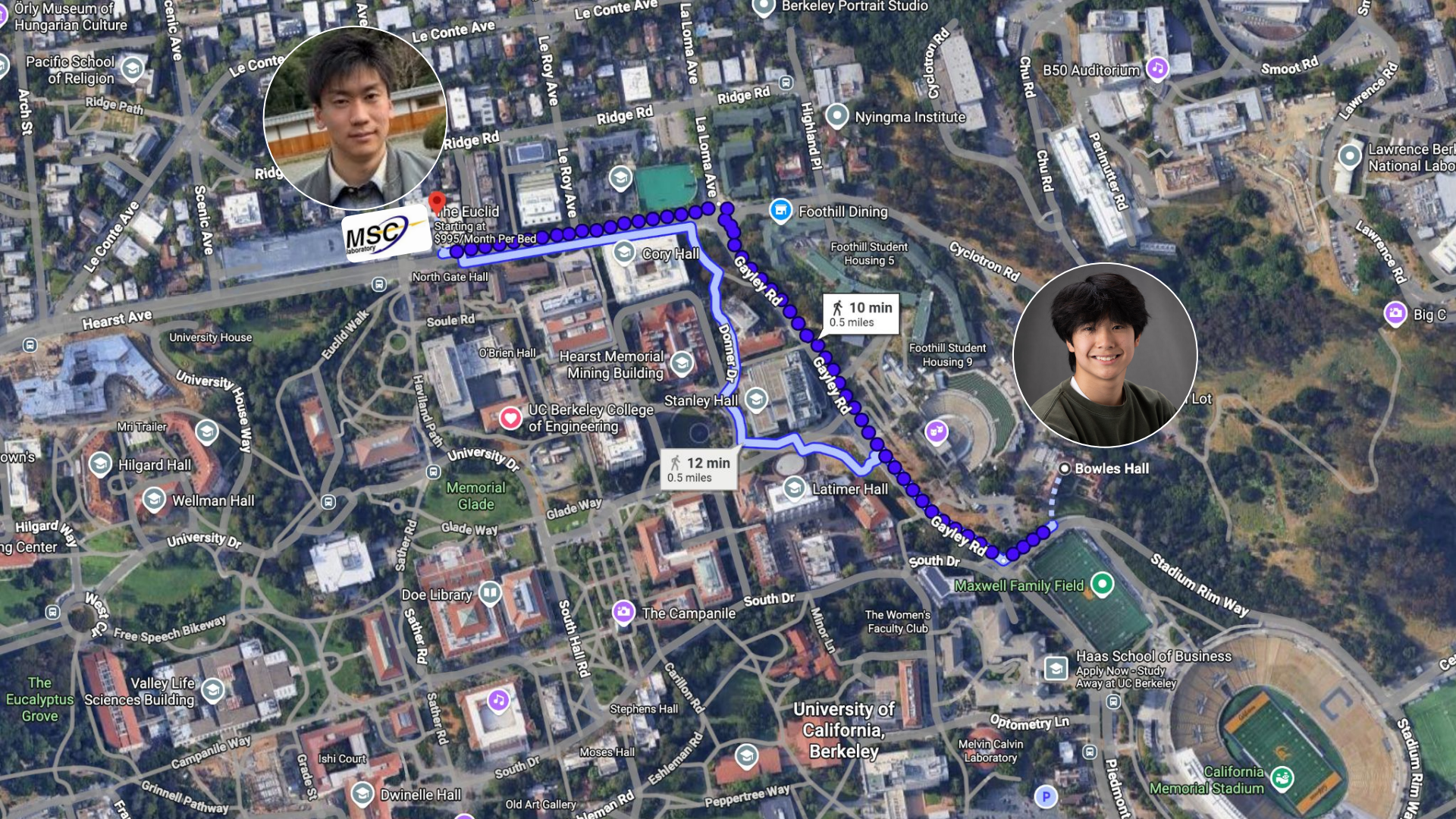
Server Rack



Watney







MSC
Laboratory

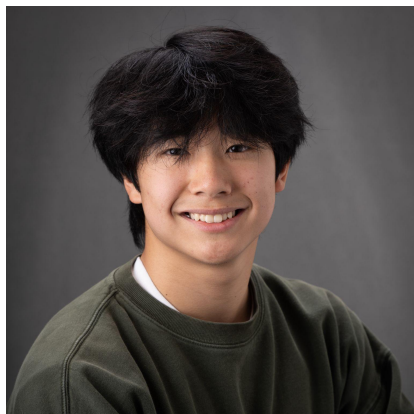
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12 min
0.5 miles

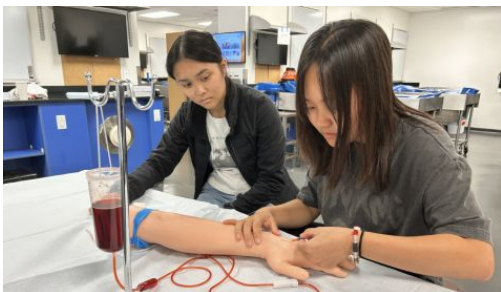
University of
California,
Berkeley

Haas School of Business
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Away at UC Berkeley



Stanford
MEDICINE

Clinical Anatomy



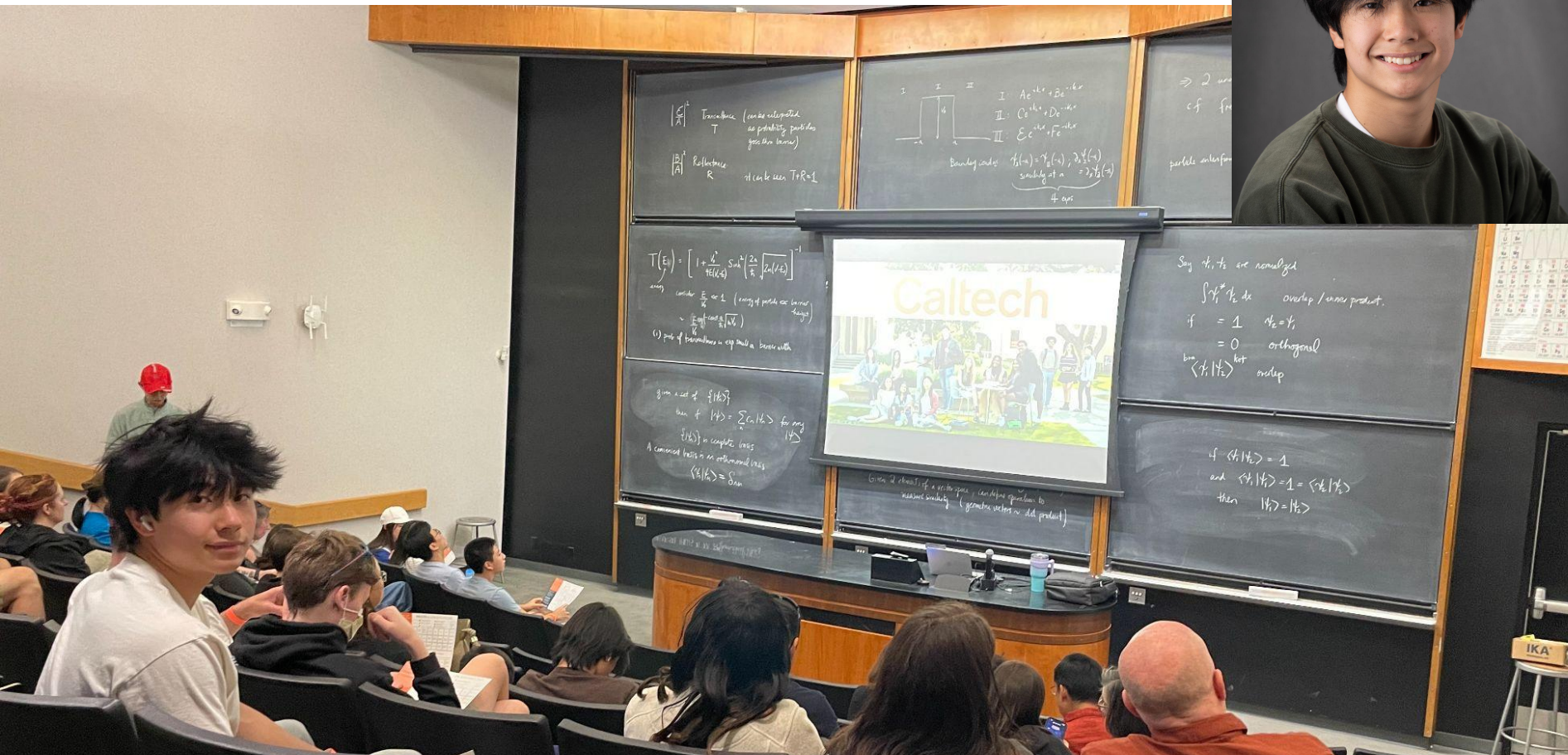
1.	 Stanford University	Biomechanical Engineering	
2.	 Massachusetts Institute of Technology	Mechanical Engineering	
3.	 Berkeley UNIVERSITY OF CALIFORNIA	Mechanical Engineering + Bioengineering + [Business]	
4.	 UNIVERSITY OF OXFORD	Engineering Science	EIT Internship
5.	 REED COLLEGE	Physics + Bioengineering	Dual Degree Program
6.	 Caltech		
7.	 JOHNS HOPKINS UNIVERSITY	Biophysics	Top Program







Caltech





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Technology



Number 1 in Mechanical
Engineering!





Stanford
University

Biomechanics Laboratory



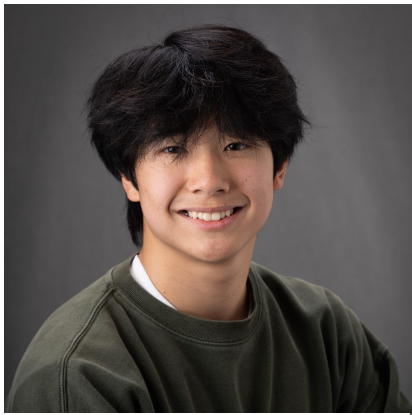
【2023~2027】



1.	Experience「経験」	Technical Skills「技術力」
A.	Watney Robotics (Silicon Valley Startup) 	Robot Arm Controller Assembly, Teleoperation, ML Models Training, PCB Soldering, CAD Design, Prototyping
B.	Tumoral Evolution (Mathematical Oncology) 	Cancer Population Dynamics, Press-Pulse, Mass Extinction, Dose Optimization, Gompertzian Growth, Fick's Laws
C.	Monte Carlo Simulation (Computational Biology) 	OpenWorm, Sibernetica, <i>C. Elegans</i> , Monte Carlo Method, Decussation, Somatic Twist, Evolutionary Biology

2.	Talented「異能」Outstanding「優れている」
a.	2 International Posters: 25th IWM & PEQG 2026
b.	5 National Contest Awards: 
c.	8 STEM Fair Prizes: 
d.	SAT Score: 1550 (760 EBRW, 790 Math)
e.	WISC-V: Extremely High (Full Scale IQ: 138)

3.	Wish to Achieve「これから成し遂げたいこと」
a.	18 Math Courses on High School Transcript (Jun 2027)
b.	Participate in iGEM Jamboree in Paris (Nov 2026)
c.	Research Paper for Regeneron STS (Jun~Oct 2026)
d.	12 Credits @ Berkeley: 3 CS/Data Courses (Aug 2026)
e.	8 AP Exams: 4 Science, 2 Math, 2 Econ (Jun 2027)



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